RUNYU CHEN

chenrunyu690@gmail.com | China

EDUCATION

Hunan University **Sep 2023 – Present**
B.Eng. in Internet of Things Engineering

RESEARCH FOCUS

Multimodal learning, perception, and control for autonomous systems (robotics and autonomous driving)

RESEARCH INTERESTS

Embodied AI (robotics and autonomous driving)
Multimodal perception and representation learning
Vision-language-action modeling

RESEARCH EXPERIENCE

CausalTAD: Injecting Causal Knowledge into Large Language Models for Tabular Anomaly Detection
Second Author, submitted to EMNLP (CCF-B),4,4,4,3.5(4,4,3,3) (https://arxiv.org/abs/2602.07798) **Jul 2025 – Present**

- Introduced a causal-aware framework to enhance tabular anomaly detection with large language models by explicitly modeling inter-column dependencies
- Developed a column reordering mechanism based on causal structure learning to preserve underlying data semantics during serialization
- Implemented causal discovery and feature reweighting modules to align model input with inferred causal importance
- Conducted extensive experiments on 30+ benchmark datasets, achieving consistent performance improvements over state-of-the-art methods

High-Precision Intelligent Anti-Pinch Control System for Automotive Power Windows
Principal Investigator, Provincial Innovation Project **Apr 2025 – Present**

- Designed a learning-based control system for automotive safety, integrating dynamic modeling and real-time feedback
- Developed an enhanced MRAC framework with improved robustness under uncertain and dynamic conditions
- Improved system stability and disturbance rejection, enabling reliable performance in complex environments
- Led a 5-member team in system design, algorithm development, and end-to-end implementation

Weakly Supervised Scene Graph Generation with Multimodal Large Models
Core Member, National Undergraduate Innovation Project **Apr 2025 – Present**

- Proposed a multimodal framework for weakly supervised scene graph generation by integrating visual and linguistic representations
- Designed and implemented a feature fusion module to capture cross-modal semantic alignment
- Developed data preprocessing pipelines and conducted comparative experiments under weak supervision settings
- Improved semantic consistency and accuracy in scene graph generation tasks

INTERNSHIP

Big Date Developer (intern), China Telecom Pingdingshan Branch **Dec 2025 – Mar 2026**

- Worked with structured enterprise datasets
- Supported SQL-based data processing in database system

AWARDS & HONORS

- Third Prize, Central-South Region Division, Chinese University Students Computer Design Competition **2024 – 2025**
- Third Prize, 27th Chinese Robotics and Artificial Intelligence Competition (Provincial-level) **2024 – 2025**
- First Prize, Zhejiang Division, 19th "Challenge Cup" Competition (Provincial Level) **2024 – 2025**
- National Encouragement Scholarship **2023 – 2024**

TECHNICAL SKILLS

- Programming: Python, C++
- Machine Learning: Transformer-based language modeling, multimodal learning, anomaly detection
- Control & Systems: Adaptive control (MRAC), dynamic system modeling